Stat 517
Solution to Homework # 1

1. For each of the following, find the constant c so that $p(x)$ satisfies the condition of being a probability mass function (pmf) of one random variable X.

   (a) $p(x) = c(\frac{2}{3})^x, x = 1, 2, 3, \ldots,$ zero elsewhere.

   (b) $p(x) = cx,\ x = 1, 2, 3, 4, 5, 6,$ zero elsewhere.

   Sol. (a)

$$\sum_{x=1}^{\infty} p(x) = c\sum_{x=1}^{\infty}(\frac{2}{3})^x = c \times 2 = 1$$

   gives $x = 1/2$.

   (b)

$$\sum_{x=1}^{6} p(x) = 21c$$

   gives $c = 1/21$.

2. If the probability density function (pdf) of X is $f(x) = 2xe^{-x^2}$, $0 < x < \infty$, zero elsewhere, determine the pdf of $Y = X^2$.

   Sol. Note that $y = x^2$ is a one-to-one transformation on $(0, \infty)$. So its inverse is $x = \sqrt{y}$ and $\frac{dx}{dy} = \frac{1}{2\sqrt{y}}$. The support of Y is $(0, \infty)$. Finally, the pdf of Y is

$$f_Y(y) = f_X(g^{-1}(y))|\frac{dx}{dy}| = e^{-y}, \quad y \in (0, \infty),$$

   where is an exponential distribution.

3. Let X have the pdf $f(x) = 3x^2$, $0 < x < 1$, zero elsewhere.

   (a) Compute $\mathbb{E}(X^3)$.

   (b) Show that $Y = X^3$ has a uniform$(0, 1)$ distribution.

   (c) Compute $E(Y)$ and compare this result with the answer obtained in part (a).

   Sol. We have

$$\begin{aligned}\mathbb{E}(X^3) &= \int_0^1 x^3 3x^2 dx \\ &= 3\int_0^1 x^5 dx = 3 \times \frac{1}{6} = \frac{1}{2}.\end{aligned}$$

If $Y = X^3$, clearly $0 < Y < 1$. For any $y \in (0,1)$,

$$\begin{aligned} F_Y(y) &= P(Y \le y) \\ &= P(X^3 \le y) \\ &= P(X \le y^{1/3}) \\ &= \int_0^{y^{1/3}} 3x^2 dx = y. \end{aligned}$$

So Y follows a uniform distribution on $(0,1)$.

Hence,

$$\mathbb{E}(Y) = \int_0^1 y \times 1 dy = \frac{1}{2},$$

which is the same as the result in part (a).

4. Let $p(x) = (\frac{1}{2})^x$, $x = 1, 2, 3, \ldots$, zero elsewhere, be the pmf of the random variable X. Find the moment generating function (mgf), the mean, and the variance of X.

   Sol. The mgf is

$$\begin{aligned} M(t) &= \mathbb{E}(e^{tX}) \\ &= \sum_{x=1}^{\infty} e^{tx} (\frac{1}{2})^x \\ &= \sum_{x=1}^{\infty} (\frac{1}{2} e^t)^x \\ &= \frac{e^t}{2 - e^t} \end{aligned}$$

   Then,

$$M'(t) = \frac{2e^t}{(2 - e^t)^2}.$$

   So $\mathbb{E}(X) = M'(0) = 2$.

   Similarly,

$$M''(t) = \frac{2e^t(2 + e^t)}{(2 - e^t)^3}.$$

   We have $\mathbb{E}(X^2) = M''(0) = 6$. So $Var(x) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = 2$.

5. Let $f(x_1, x_2) = 4x_1x_2$, $0 < x_1 < 1$, $0 < x_2 < 1$, zero elsewhere, be the pdf of X_1 and X_2. Find $P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1)$, $P(X_1 = X_2)$, $P(X_1 < X_2)$, and $P(X_1 \le X_2)$.

   Sol.

$$\begin{aligned} P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1) &= \int_{1/4}^1 \int_0^{1/2} 4x_1x_2 dx_1 dx_2 \\ &= \frac{1}{2} \int_{1/4}^1 x_2 dx_2 \\ &= \frac{15}{64}. \end{aligned}$$

$P(X_1 = X_2) = 0$ because there is no volume over $(x_1, x_2) : x_1 = x_2$.

$$\begin{aligned} P(X_1 < X_2) &= \int_0^1 \int_0^{x_2} 4x_1x_2 dx_1 dx_2 \\ &= \frac{1}{2}. \end{aligned}$$

$$P(X_1 \leq X_2) = P(X_1 = X_2) + P(X_1 < X_2) = \frac{1}{2}.$$

6. Let $f(x, y) = e^{-x-y}$, $0 < x < \infty$, $0 < y < \infty$, zero elsewhere, be the pdf of X and Y. Then if $Z = X + Y$, compute $P(X \leq 0)$, $P(Z \leq 6)$, and more generally $P(Z \leq z)$ for $0 < z < \infty$. What is the pdf of Z?

   Sol. For $0 < z < \infty$,

$$\begin{aligned} P(Z \leq z) &= \int_0^z \int_0^{z-x} e^{-x-y} dy dx \\ &= \int_0^z (-e^{-z} + e^{-x}) dx \\ &= 1 - e^{-z} - ze^{-z}. \end{aligned}$$

   $P(Z \leq 0) = 0$ because Z is a positive random variable.

   $P(Z \leq 6) = 1 - e^{-6} - 6e^{-6} = 0.98.$

   The pdf of Z is $f_Z(z) = F'_Z(z) = ze^{-z}$ for $z > 0$.

7. Let X_1 and X_2 have the joint pdf $f(x_1, x_2) = 15x_1^2x_2$, $0 < x_1 < x_2 < 1$, zero elsewhere. Find the marginal pdfs and compute $P(X_1 + X_2 \leq 1)$.

   Sol. The marginal pdfs are:

$$f_1(x_1) = \int_{x_1}^1 15x_1^2x_2 dx_2 = 15x_1^2(\frac{1}{2} - \frac{1}{2}x_1^2), \qquad 0 < x_1 < 1,$$

$$f_2(x_2) = \int_0^{x_2} 15x_1^2x_2 dx_2 = 5x_2^4, \qquad 0 < x_2 < 1.$$

   Try to graph the space X_1 and X_2 and carefully choose the limits of integration for the joint probability.

$$\begin{aligned} P(X_1 + X_2 \leq 1) &= \int_0^{1/2} \int_{x_1}^{1-x_1} 15x_1^2x_2 dx_2 dx_1 \\ &= \int_0^{1/2} 15x_1^2(\frac{1-2x_1}{2}) dx_1 \\ &= 0.078. \end{aligned}$$

8. Let X and Y have the joint pdf $f(x,y) = 3x$, $0 < y < x < 1$, zero elsewhere. Are X and Y independent? If not, find $E(X|y)$.

Sol. We first get the marginal pdfs.

$$f_X(x) = \int_0^x 3x dy = 3x^2, \quad 0 < x < 1.$$

$$f_Y(y) = \int_y^1 3x dx = \frac{3}{2}(1 - y^2), \quad 0 < y < 1.$$

Note that $f(x,y) \neq f_X(x) f_Y(y)$. So they are not independent.

$$E(X|Y = y) = \int_y^1 x \frac{3x^2}{3/2(1 - y^2)} dx = \frac{2(1 - y^3)}{3(1 - y^2)}.$$